

Project Title

IPL Performance Analytics

Project Objective

The objective of this project is to analyze IPL match and delivery data, identify team and player performance trends, and present the findings using Python visualizations and a Power BI dashboard.

Dataset Description

The project uses IPL matches and deliveries datasets. The data contains match details, teams, players, venues, runs, wickets, toss decisions, and match results.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Microsoft Power BI.

Data Cleaning

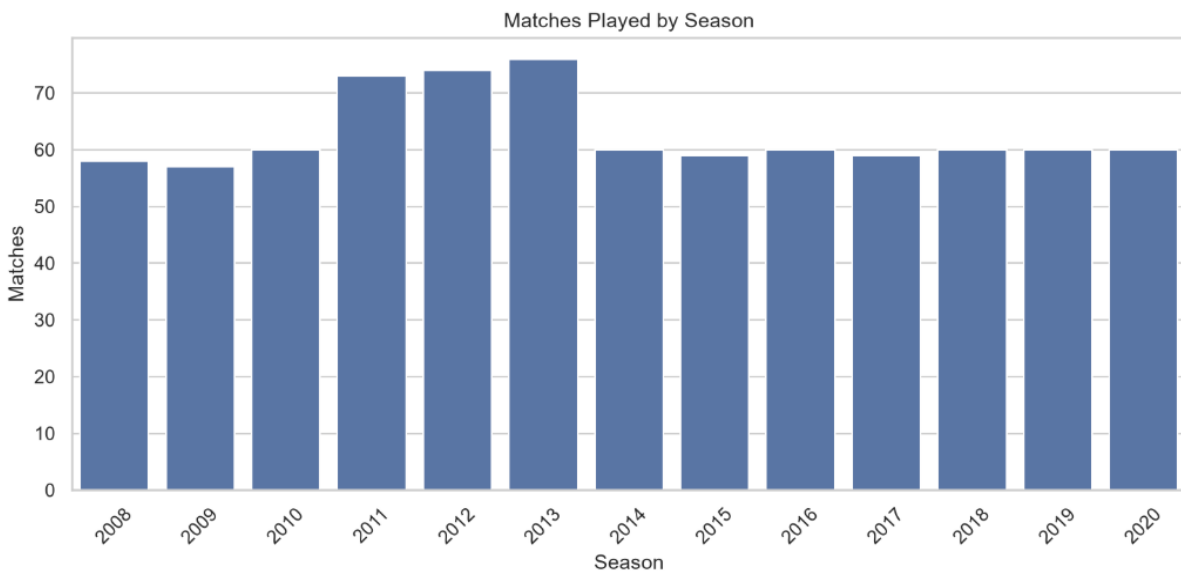
The datasets were loaded and inspected for missing or incorrect values. Column names and data types were checked, date values were formatted where required, and the cleaned datasets were saved for analysis.

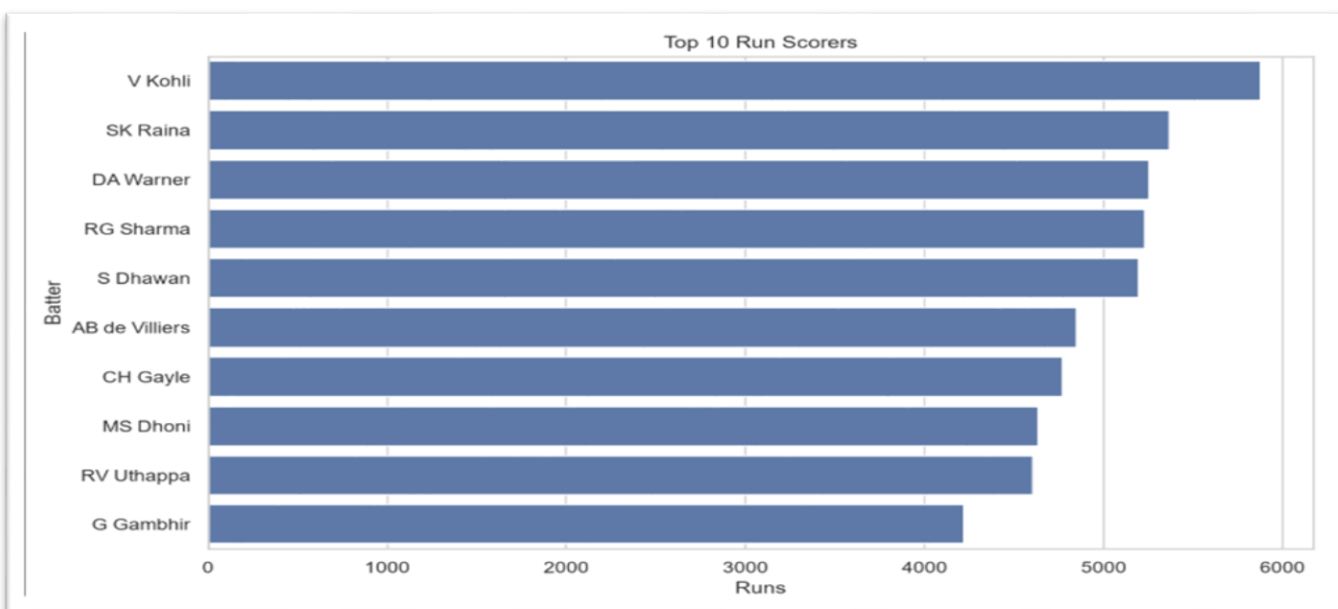
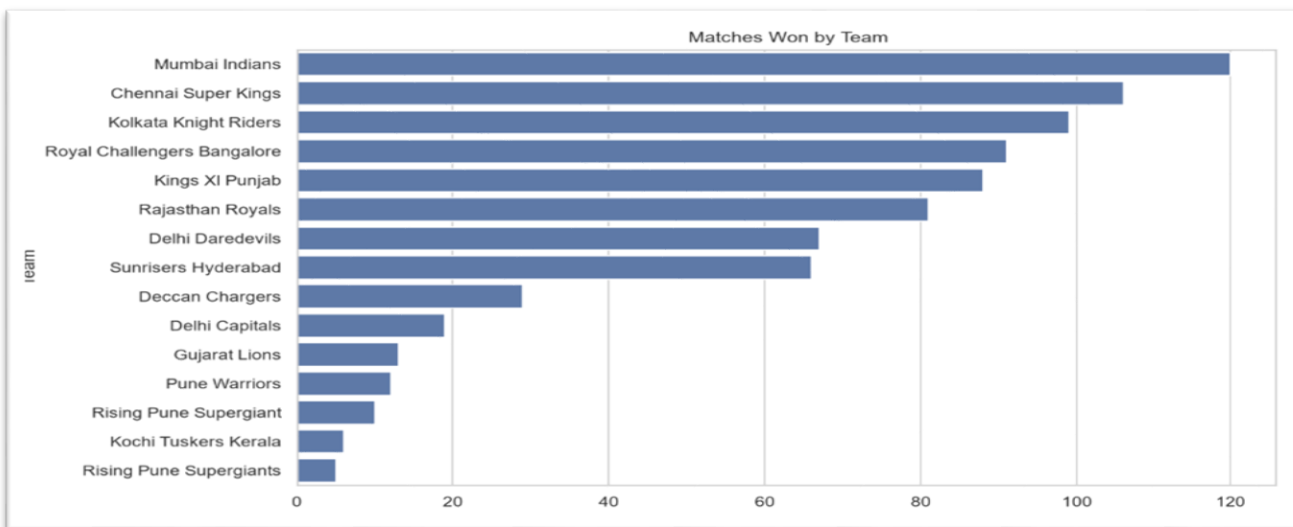
Data Analysis

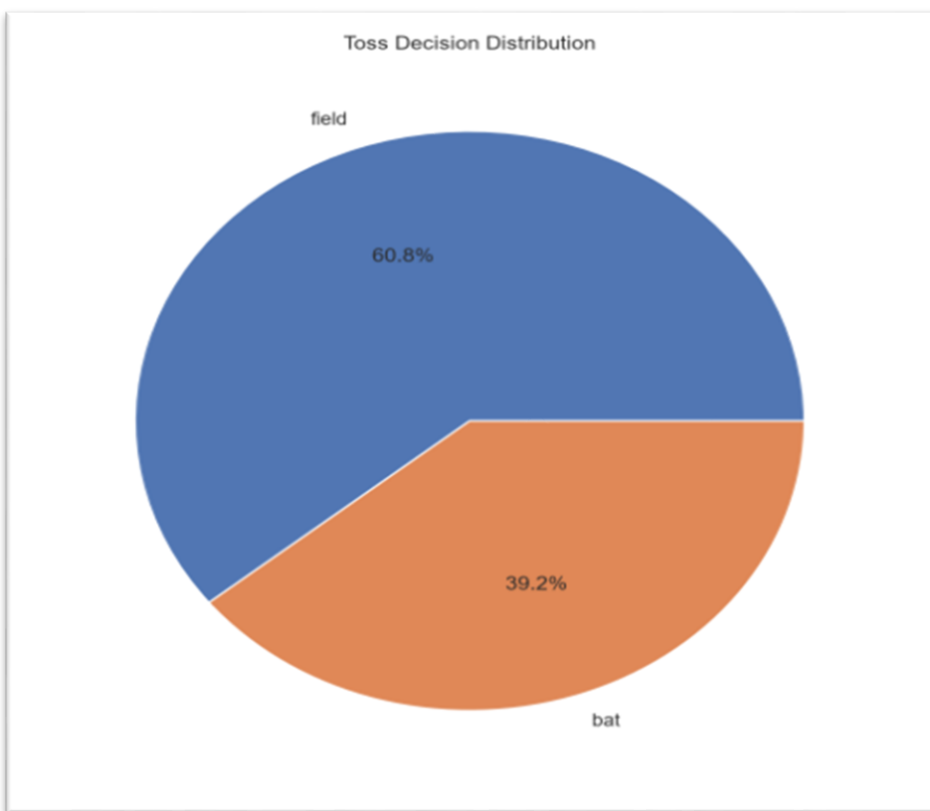
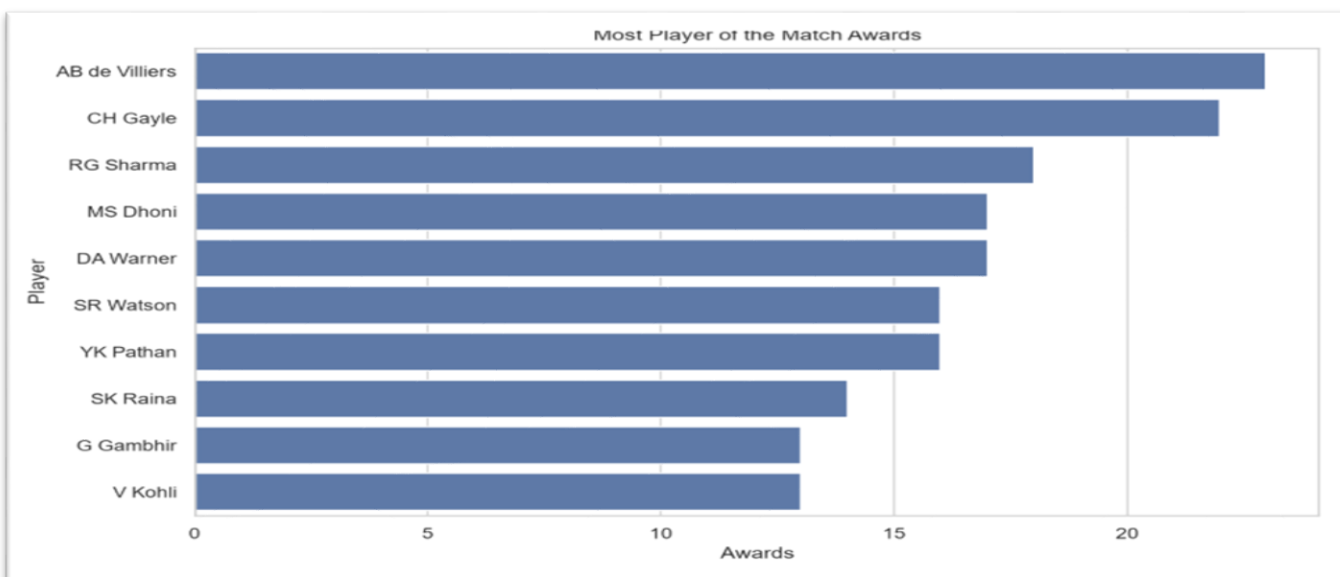
The cleaned data was analyzed to calculate match totals, team wins, player runs, wickets, toss decisions, venue counts, and season-wise performance.

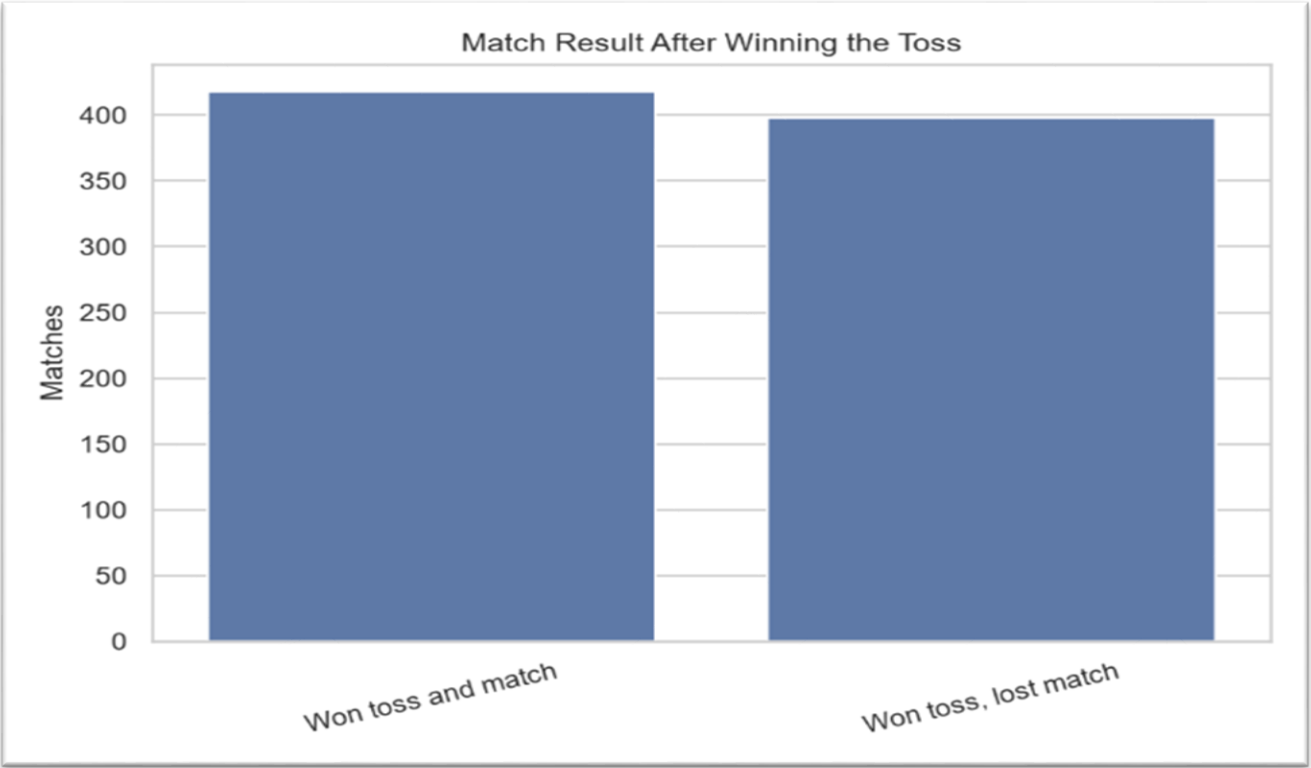
Python Visualizations

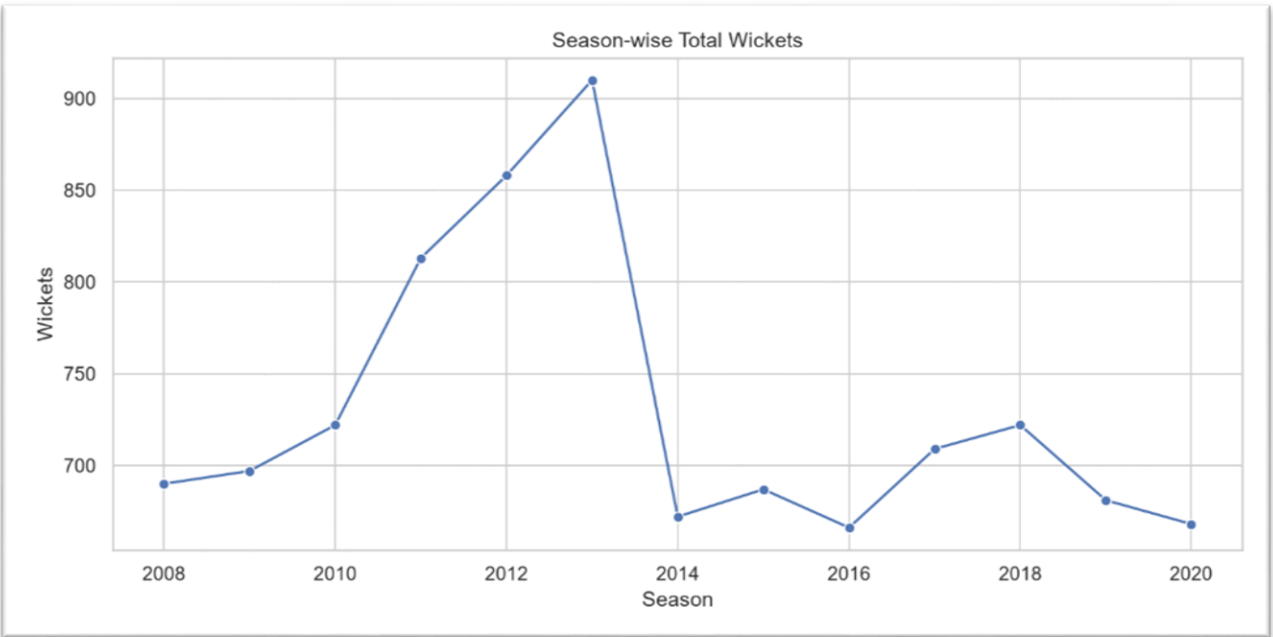
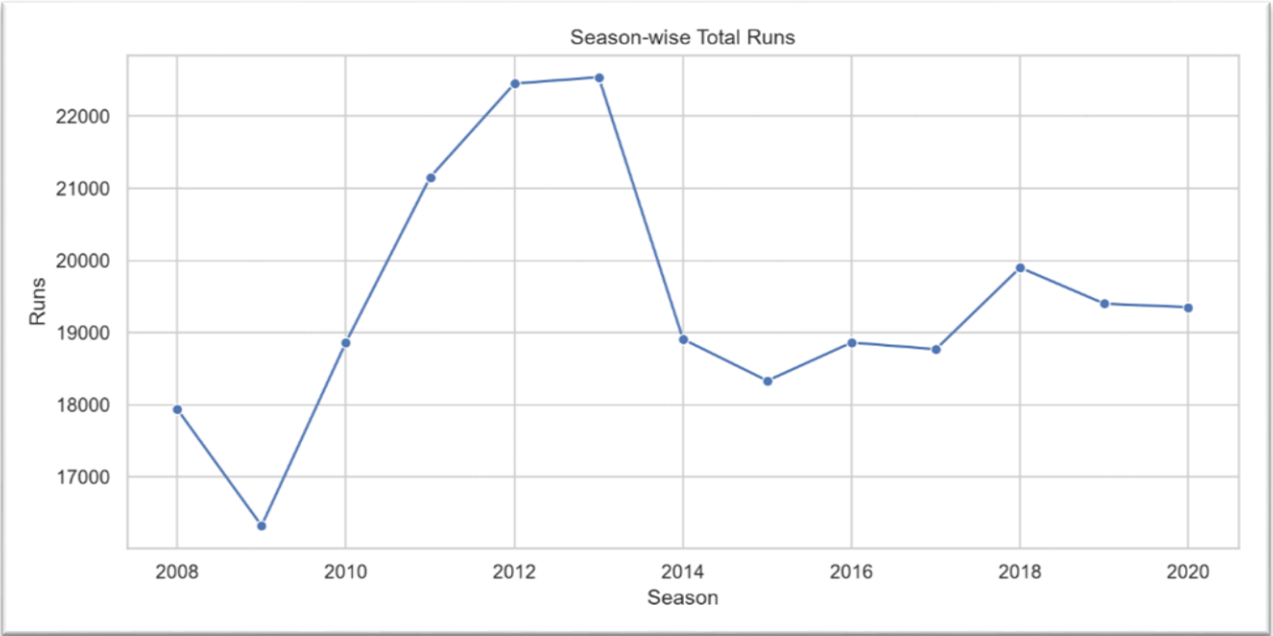
Python visualizations were created to show season-wise matches and runs, team wins, top batsmen and bowlers, toss decisions, venue counts, run distributions, and correlations.

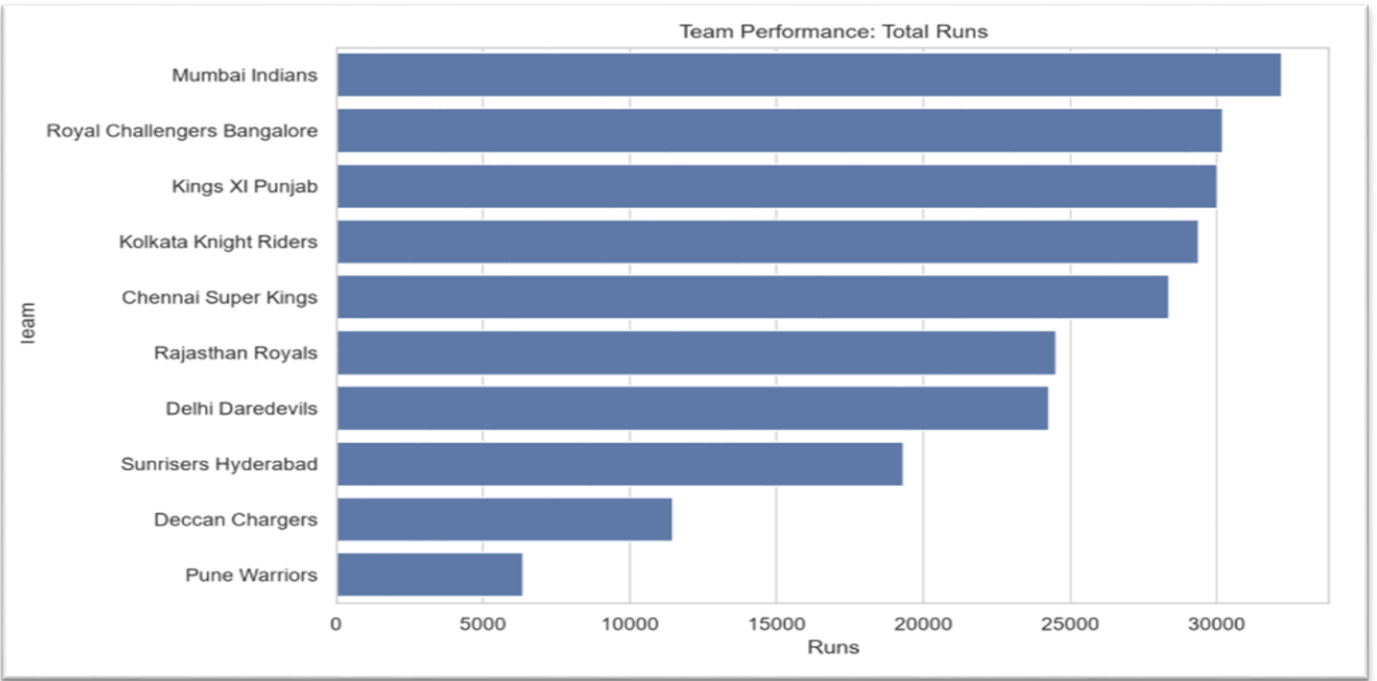
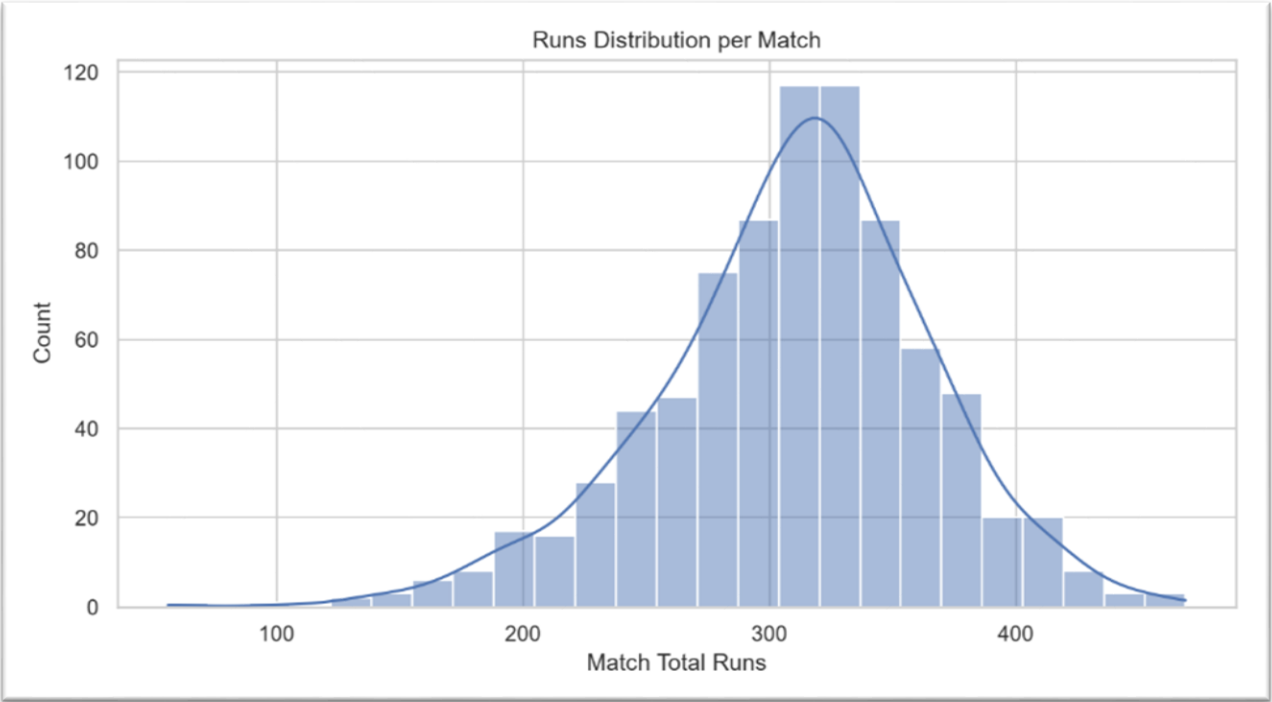


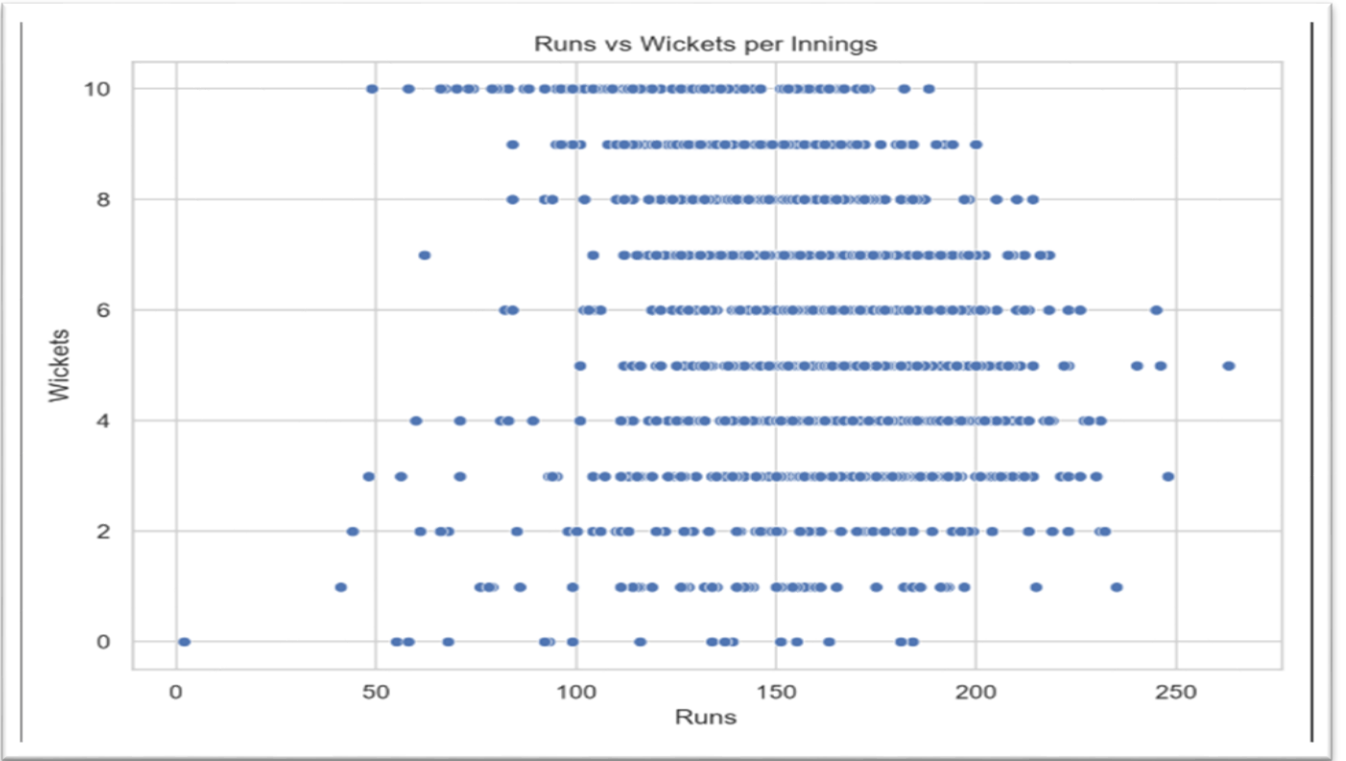
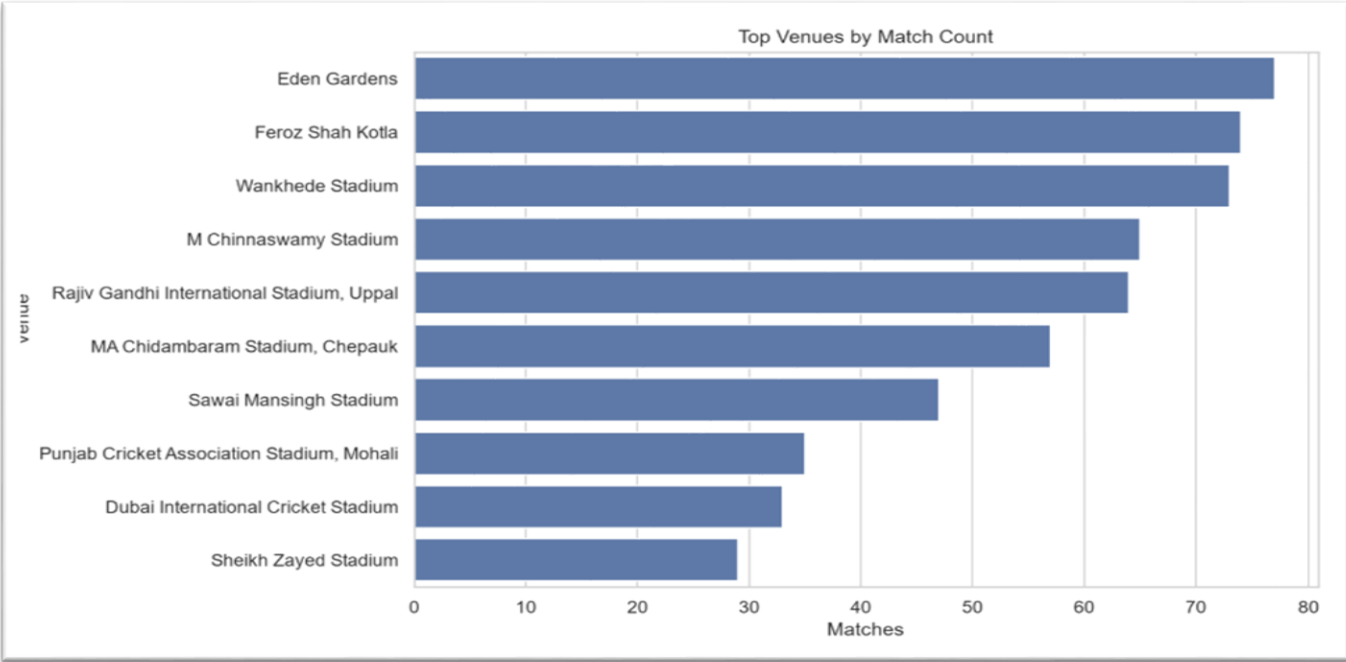


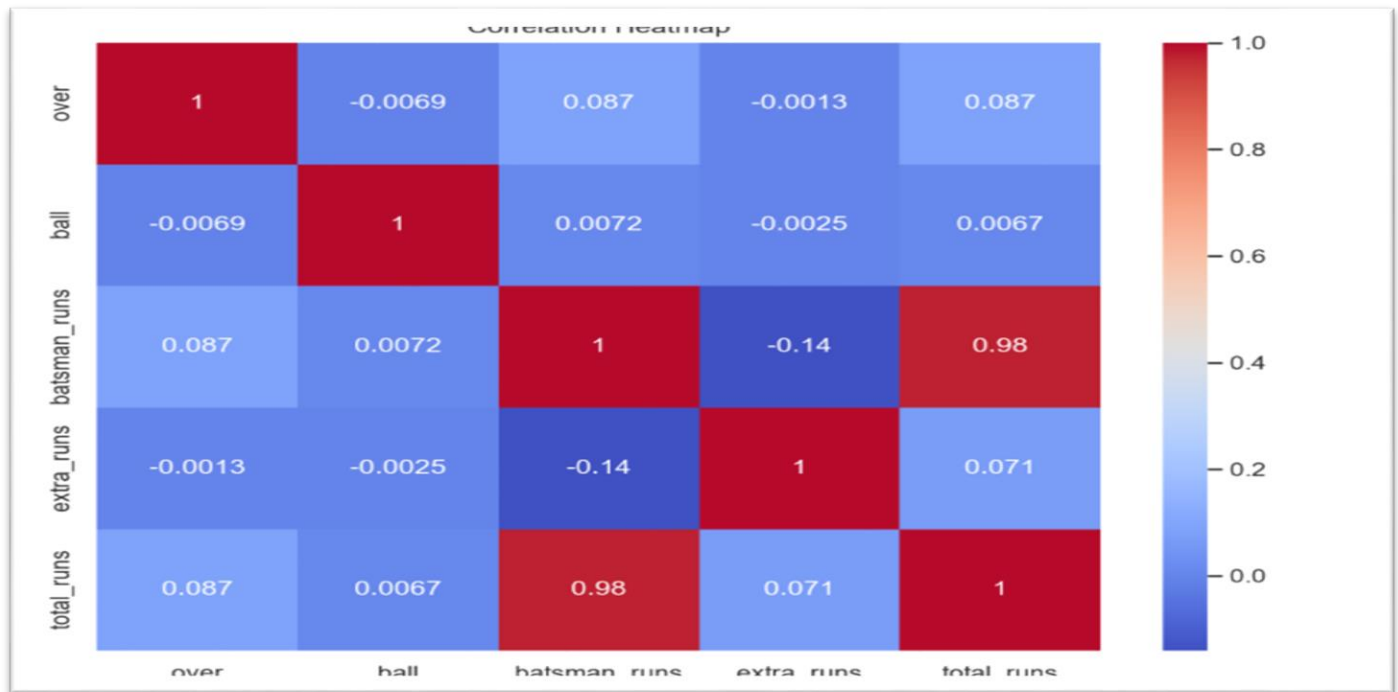












Power BI Dashboard Screenshots

An interactive IPL Performance Analytics Dashboard was created with KPI cards, charts, and season, team, and venue filters.

IPL Performance Analytics Dashboard

KPI CARDS

Total Matches
816

Total Runs
253K

Total Wickets
9K

Total Teams
15

Total Players
580

Total Wins
816

Average Runs
309.80

Average Winning Margin
17.32

Total Deliveries
193K

Total Delivery Rows
193K

Win Percentage
100.00%

Slicers

Season filter



Team Filter

- team1
- ☐ Chennai Super Kings
 - ☒ Deccan Chargers
 - ☐ Delhi Capitals
 - ☐ Delhi Daredevils
 - ☐ Gujarat Lions
 - ☐ Kolkata Knight Riders

Venu Filter

- venue
- ☐ Barabati Stadium
 - ☐ De Beers Diamond Oval
 - ☐ Dr DY Patil Sports Academy
 - ☐ Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium
 - ☐ Kingsmead
 - ☐ M. Chinnappa Reddy

Team Performance

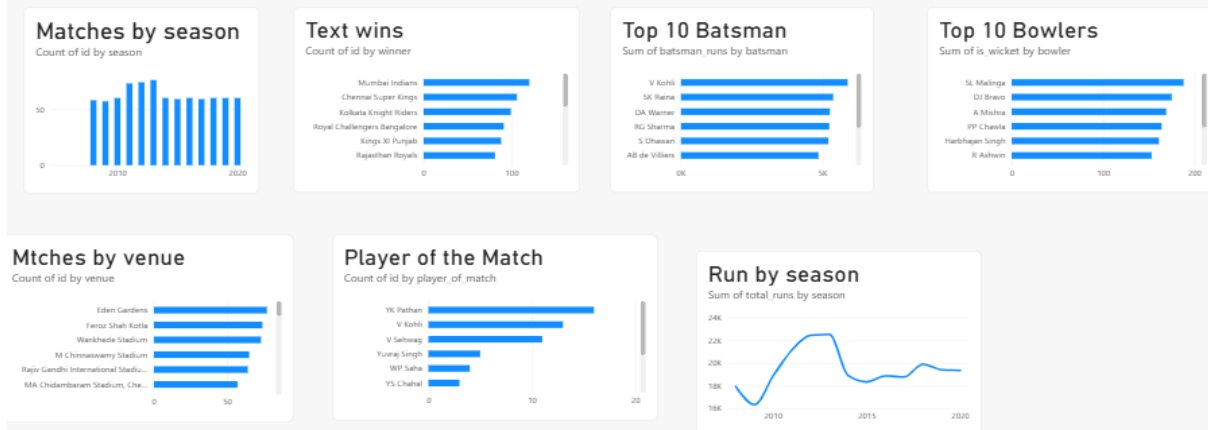
winner	Count of id
Chennai Super Kings	2
Deccan Chargers	13
Delhi Daredevils	5
Kings XI Punjab	5
Kolkata Knight Riders	4
Total	39

Toss Decision

Count of id by toss, decision



Charts



Key Insights

- Mumbai Indians recorded the highest number of wins, with 120 wins.
- The 2013 season had the highest number of matches, with 76 matches.
- The 2013 season recorded the highest total runs, with 22,541 runs.
- Virat Kohli was the leading run scorer in the batsmen chart.
- Lasith Malinga was the leading wicket taker in the bowlers chart.
- Eden Gardens hosted the highest number of matches in the venue chart.
- Toss winners won 51.48% of matches where both the toss winner and match winner were known

Conclusion

This project used Python and Power BI to analyze IPL data and present team, player, and season performance in clear visualizations and an interactive dashboard.